DC Generator Video: https://youtu.be/mq2zjmS8UMI?si=W6HD7kTGB3BiQ8-M
DC Motor Video Link: https://youtu.be/j_F4limaHYI?si=8z9eP3VPfwBgXJkJ

Commutation

- Commutation is the process of converting the AC voltages and currents in the rotor of a DC machine to DC voltages and currents at its terminals
- When the voltage reverses in a loop, the connections of the loop are also switched to keep the polarity of the terminal voltage the same
- Increasing the number of loops on the rotor, we can improve our approximation to perfect DC voltage
- Commutator segments are usually made out of copper bars and the brushes are made of a mixture containing graphite to minimize friction between segments and brushes

Example 26.3. *A shunt generator delivers 450 A at 230 V and the resistance of the shunt field and armature are 50 Ω and 0.03 Ω respectively. Calculate the generated e.m.f.*

Solution. Generator circuit is shown in Fig. 26.46.

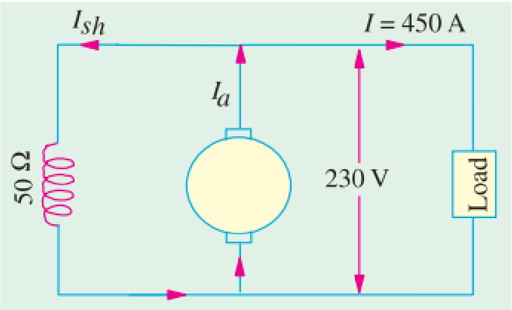

Fig. 26.46

Current through shunt field winding is

$$I_{sh} = 230/50 = 4.6 \text{ A}$$

Load current $\quad I = 450$ A

$\therefore$ Armature current $I_a = I + I_{sh}$

$$= 450 + 4.6 = 454.6 \text{ A}$$

Armature voltage drop

$$I_aR_a = 454.6 \times 0.03 = \mathbf{13.6\ V}$$

Now $\quad E_g$ = terminal voltage + armature drop

$$= V + I_aR_a$$

$\therefore$ e.m.f. generated in the armature

$$E_g = 230 + 13.6 = \mathbf{243.6\ V}$$

Example 29.21. *A 250-V shunt motor runs at 1000 r.p.m. at no-load and takes 8A. The total armature and shunt field resistances are respectively 0.2 Ω and 250 Ω. Calculate the speed when loaded and taking 50 A. Assume the flux to be constant.* **(Elect. Engg. A.M.Ae. S.I. June 1991)**

Solution. Formula used : $\dfrac{N}{N_0} = \dfrac{E_b}{E_{b0}} \times \dfrac{\Phi_0}{\Phi}$; Since $\Phi_0 = \Phi$ (given); $\dfrac{N}{N_0} = \dfrac{E_b}{E_{b0}}$

$$I_{sh} = 250/250 = 1 \text{ A}$$

$$E_{b0} = V - I_{a0}R_a = 250 - (7 \times 0.2) = 248.6 \text{ V}; \; E_b = V - I_aR_a = 250 - (49 \times 0.2) = 240.2 \text{ V}$$

$$\therefore \quad \frac{N}{1000} = \frac{240.2}{248.6}; \quad N = \mathbf{9666.1\ r.p.m.}$$